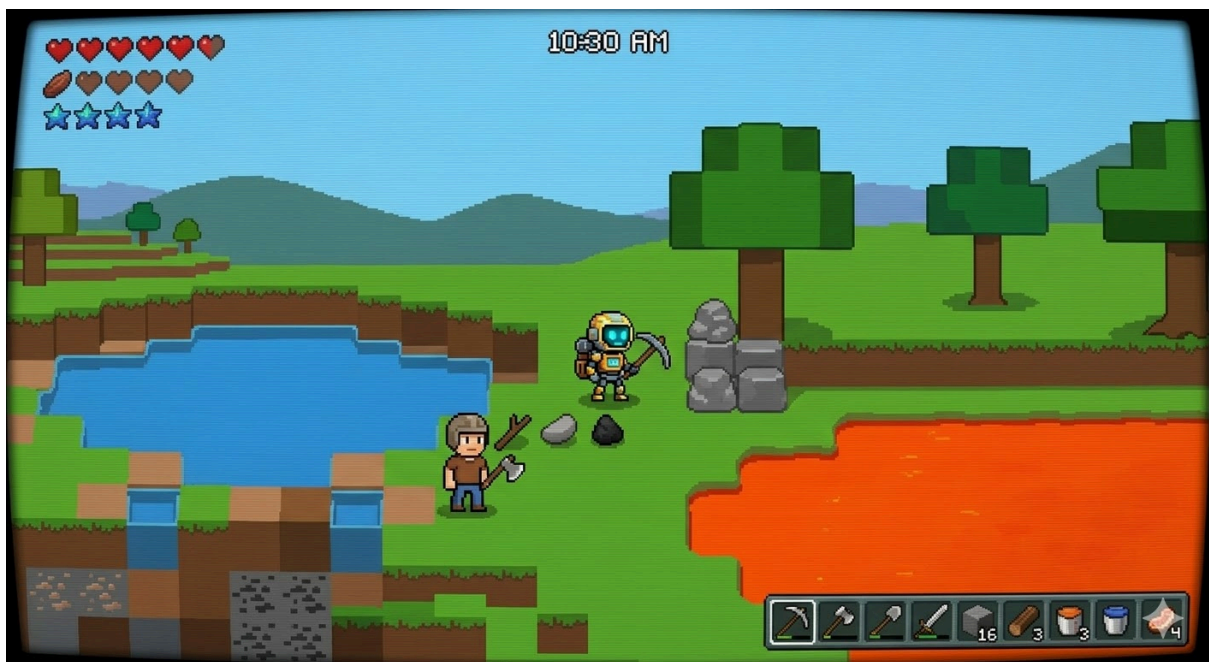


Autonomous Agents Assignment Proposal

For this project I have to have a “lifeform” that is able to survive, interact with the environment, interact with the player, and have cute eyes.

My idea is to make a maybe somewhat traditional seeming game but it contains a lifeform whose entire life goal is to annoy you. This lifeform is going to be called B.O.B. The idea is to start off with a survival style game where a player spawns into the world, they start off by collecting basic survival materials such as wood, stone, food, fuel, etc. While they are collecting these resources there will be the B.O.B following them the whole time making their life as difficult as possible. B.O.B can act like the player too in the sense that he will be able to collect resources and create tools and such just as the player can but B.O.B's life goal is to annoy the player so he uses these materials and tools to sabotage the players progress. He could punch the player to disrupt the players mining, he could break the players chest to destroy his stuff, he could close all the doors that a player opens, etc.



The project will be built as a 2D game in Godot 4 with a pixel art art style. For the autonomous nature of the robot it will function along these lines:

B.O.B's Needs:

- Hunger: B.O.B will have a hunger bar the same as the player that when empty will cause him to lose health
- Safety: B.O.B can be damaged by the environment so he will need to do his best to stay safe of danger such as lava or monsters.
- Curiosity: B.O.B will also have curiosity which will encourage him to spend some time away from the player to gather new resources and explore new areas. This could lead to B.O.B dying so he will have **caution** element too where he becomes aware of possible dangers.

Some internal flags could be as follows:

- **is_night**
- **has_shelter**
- **in_danger**
- **has_tools**

B.O.B will move around using steering behaviour avoiding obstacles and taking direct lines to his task where it fits. He will follow the same interactional behaviour as the player; Harvesting/mining, building, and consuming/eating. All of the interactions including movement will have animations to indicate what is being done.

The environment will be handcrafted (if time allows procedural generation may be added) with the addition of having certain entities spawn in randomly around the world like monsters or animals. There will also be a day/night cycle which affects the behaviour of the monsters and animals in both spawning and existing.

Also side note, B.O.B stands for **Built On Bullying** since his life goal to to annoy/"bully" the player 👍